

De novo H3.3K27M-altered Diffuse Midline Glioma in human brainstem organoids to dissect GD2 CAR T cell function

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This manuscript has been previously reviewed at another journal. This document only contains information relating to versions considered at Nature Cancer.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have addressed all prior comments. This study is important in establishing pontine organoids and advances the field of DMG research.

Reviewer #2

(Remarks to the Author)

The authors have addressed my questions. I acknowledge that the interactions between microglia and GD2 CAR-T cells present some novel aspects. However, the broader interaction between microglia and CAR-T cells has been explored in previous studies. For example:

ref.83, Targeting TGF β -activated kinase-1 activation in microglia reduces CAR T cell-associated neurotoxicity syndrome.

Immunotherapy-related cognitive impairment following CAR T cell therapy in mice.

That said, this is not the primary focus of the current study, and additional experiments are necessary to conclusively determine the functional role of microglia in CAR-T cell activity. For instance, using other cell types as controls would help confirm that the observed effects are specific to microglia.

Reviewer #3

(Remarks to the Author)

The authors of De novo H3.3K27M-altered Diffuse Midline Glioma in human brainstem organoids to dissect GD2 CAR T cell function present an elegant FGF4-driven human brainstem organoid model of DMG. By co-culturing these organoids with myeloid cells, they demonstrate—consistent with patient data—that GD2-targeting CAR T cells provide only limited therapeutic benefit.

The Introduction is strong, but could be strengthened by including a brief discussion of the role the tumour microenvironment plays in DMG, particularly neuronal crosstalk. While this is addressed later in the Results, mentioning it earlier would help to frame the study's significance and provide better context.

Investigations of cellular composition show region-specific identities emerging over time, including a strong hindbrain and pronounced pontine signature. Cells transition from a pluripotent state to neuroepithelial, radial glial, and eventually into distinct neuronal and glial populations. Could the authors align this time-course data with human developmental timelines to estimate which developmental stage corresponds to the median age of DMG diagnosis, when H3.3K27M, TP53, and PDGFRA alterations are most commonly found? Given that DMG onset and survival are closely tied to age-specific cellular context, this alignment could add further translational value.

DEG should be expanded at first use.

As mentioned above, a limitation of the study is the lack of functional characterisation between the DMGO and the surrounding microenvironment. This is particularly relevant in light of recent findings showing physical interactions between nerves and tumour cells, where cancer-associated neurons enhance tumour metabolic plasticity by transferring mitochondria to cancer cells. Some discussion or future consideration would strengthen the manuscript's translational framing.

The dependence of the BrOs on a gliogenic state for DMG transformation is intriguing. It is notable that H3K27M plasmids were not retained in unguided cerebral organoids, whereas TP53 (DNp53) was. Can the authors comment on the behaviour of PDGFRA in this context? Additionally, could they speculate on whether factors expressed in unguided organoids might inhibit DMG transformation, and whether such factors could inform differentiation-based therapeutic approaches?

The CAR T cell characterisation post-DMGO exposure is elegant and informative. However, a limitation is the lack of deeper analysis of resistant DMGOs post-treatment, particularly in the context of co-culture with microglia, where immunosuppressive transcriptional profiles would be expected. A more detailed exploration here would improve the robustness of the conclusions.

Finally, in NCAM+ T cells, can the authors identify exhaustion markers, such as PI3K pathway overactivation or epigenetic alterations, that may represent potential intervention points for small-molecule therapies—particularly those already used during CAR T cell manufacturing?

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Reviewer #1:

Remarks to the Author:

The authors have addressed all prior comments. This study is important in establishing pontine organoids and advances the field of DMG research.

We are very grateful for the constructive feedback provided throughout the review process and are pleased that the reviewer deems our work establishing a pontine-regionalized brain organoid model important and advancing DMG research.

Reviewer #2:

Remarks to the Author:

The authors have addressed my questions. I acknowledge that the interactions between microglia and GD2 CAR-T cells present some novel aspects. However, the broader interaction between microglia and CAR-T cells has been explored in previous studies. For example:

ref.83, Targeting TGF β -activated kinase-1 activation in microglia reduces CAR T cell-associated neurotoxicity syndrome.

Immunotherapy-related cognitive impairment following CAR T cell therapy in mice.

That said, this is not the primary focus of the current study, and additional experiments are necessary to conclusively determine the functional role of microglia in CAR-T cell activity. For instance, using other cell types as controls would help confirm that the observed effects are specific to microglia.

We strongly agree with the reviewer about the importance of this previous work (*ref. 83*), as well as the most recent study published when our revised manuscript was under consideration (*Gerathy et al.*, *Cell*, June 2025). While these studies experimentally address – in mouse models - how microglia contribute to CAR T cell toxicity, as well as offering correlative patient data, our work primarily focused on the impact of microglia on CAR T cell efficacy. An increase in pro-inflammatory cytokines in the presence of microglia (**Fig. 6b**) aligns with this previous work and suggest that this model could also potentially be used to test toxicity mitigation strategies. However, importantly we were able to demonstrate a reduction in tumor control by CAR T cells in the presence of microglia that to our knowledge has not been addressed experimentally, only suggested through correlative clinical trial data (*Majzner et al.*, *Nature*, 2022). We agree with the reviewer that additional experiments are needed to establish the extend of this effect also taking into consideration heterogeneity of outcomes that are captured with our DMG organoid (DMGO) model. In addition, the underlying mechanisms of this CAR T cell microglia interplay - as also acknowledged in our Discussion on page 20-21, lines 459-462 - await further investigation; with the identification of direct microglia and CAR T cell interactions (**Fig. 6a**) and the functional GD2 CAR T cell profiles susceptible to microglia (**Fig. 6c**) offering a critical starting point for this line of work.

Using other cell types as controls in these experiments will indeed help to demonstrate microglia specificity of the observed effects. However, integrating the appropriate control cells will not be straightforward, as we showed that primitive macrophage progenitors (PMPs) spontaneously differentiate into microglia in the microenvironmental context of brainstem-regionalized organoids and adopt DMG-associated states in DMGOs (Fig. 5).

Reviewer #3:

Remarks to the Author:

The authors of De novo H3.3K27M-altered Diffuse Midline Glioma in human brainstem organoids to dissect GD2 CAR T cell function present an elegant FGF4-driven human brainstem organoid model of DMG. By co-culturing these organoids with myeloid cells, they demonstrate—consistent with patient data—that GD2-targeting CAR T cells provide only limited therapeutic benefit.

We are pleased that the reviewer finds our new protocol to generate brainstem-regionalized organoids elegant and agrees that DMG modeled in these organoids recapitulates CAR T cell treatment outcomes observed in patients.

The Introduction is strong, but could be strengthened by including a brief discussion of the role the tumour microenvironment plays in DMG, particularly neuronal crosstalk. While this is addressed later in the Results, mentioning it earlier would help to frame the study's significance and provide better context.

As suggested, we now introduce this specific importance of the unique microenvironment; neuronal interplay, in the introduction of our revised manuscript on page 3, lines 66-70.

Investigations of cellular composition show region-specific identities emerging over time, including a strong hindbrain and pronounced pontine signature. Cells transition from a pluripotent state to neuroepithelial, radial glial, and eventually into distinct neuronal and glial populations. Could the authors align this time-course data with human developmental timelines to estimate which developmental stage corresponds to the median age of DMG diagnosis, when H3.3K27M, TP53, and PDGFRA alterations are most commonly found? Given that DMG onset and survival are closely tied to age-specific cellular context, this alignment could add further translational value.

We acknowledge the value of *in vitro* models that closely mimic the age of diagnosis as a valuable tool for clinical testing. However, it has been shown that human embryonic stem cell (hESC)-derived neural organoids resemble more closely first and second trimester of human fetal development (He *et al.*, Nature, 2024). It should be noted that these alignments are done in telencephalic neural organoid systems, which could differ from the faster developing hindbrain (Braun *et al.*, Science, 2023). However, to our current knowledge, there

is no comprehensive atlas covering early human post-natal development of the developing hindbrain or midline to date that would enable such an alignment in our hindbrain organoid system.

We also acknowledge this in the Discussion of our manuscript (page 19, lines 431-433), stating that, as our model is human embryonic stem cell (hESC)-derived, it reflects a developmental stage equivalent to the second trimester, as reported for other neural organoids (Gordon *et al.*, Nature Neuroscience, 2021; He *et al.*, Nature, 2024). While this developmental window for inducing DMG tumorigenesis aligns with murine (Pathania *et al.*, Cancer Cell, 2017; Patel *et al.*, Neuro-Oncology, 2019; Larson *et al.*, Cancer Cell, 2019; Tomita *et al.*, Glia, 2022) and *in vitro* studies (Funato *et al.*, Science, 2014; Haag *et al.*, Cancer Cell, 2021; Bressan *et al.*, Cell Stem Cell, 2021) that point to an early embryonic period in which NSCs are most permissive to H3.3K27M-driven transformation, this will of course be challenging to confirm in patients. In mice it has been shown that H3.3K27M also retains tumorigenic potential beyond postnatal gliogenesis (P0/P1) (Larson *et al.*, Cancer Cell, 2019), thus the time of transformation and how it reflects to onset of symptoms in patients remains elusive.

DEG should be expanded at first use.

We thank the reviewer for noticing this oversight and have now included this on page 7, line 156 of our revised manuscript.

As mentioned above, a limitation of the study is the lack of functional characterisation between the DMGO and the surrounding microenvironment. This is particularly relevant in light of recent findings showing physical interactions between nerves and tumour cells, where cancer-associated neurons enhance tumour metabolic plasticity by transferring mitochondria to cancer cells. Some discussion or future consideration would strengthen the manuscript's translational framing.

We agree with the reviewer that further and experimental characterization of such tumor-neuronal crosstalk will support model validation for these key processes involved in tumor progression. We now mention this required future direction (that we are actively pursuing in ongoing projects in the lab) in the Discussion of our revised manuscript (page 20, lines 438-442). However, in the current study our main focus was on understanding CAR T cell biology and enhancing their efficacy in the context of this neuronal - as well as the brain-resident immune - microenvironment.

The dependence of the BrOs on a gliogenic state for DMG transformation is intriguing. It is notable that H3K27M plasmids were not retained in unguided cerebral organoids, whereas TP53 (DNp53) was. Can the authors comment on the behaviour of PDGFRA in this context? Additionally, could they speculate on whether factors expressed in unguided organoids

might inhibit DMG transformation, and whether such factors could inform differentiation-based therapeutic approaches?

As noted in the text, previous attempts to generate H3.3K27M-altered DMG tumors in unguided cerebral organoids have not been successful (Bian *et al.*, Nature Methods, 2018). Although we do not have PDGFRA data from our unguided organoids, others have shown that introducing this amplification alone does not result in tumor formation. The same holds true for P53 or H3.3K27M individually. However, P53 in combination with other mutations can promote tumorigenesis, yet not with H3.3K27M. For instance, even the combined introduction of P53, H3.3K27M, and ATRX loss (another recurrent mutation in DMG; see Schwartzenruber *et al.*, Nature, 2012; Mackay *et al.*, Cancer Cell, 2017) failed to induce tumors in cerebral organoids (Bian *et al.*, Nature Methods, 2018), underscoring that H3.3K27M requires the appropriate cellular context to drive DMG formation.

In our case, we observed only very few tumors. In the tumors large enough to image at high resolution, we did not detect H3.3K27M expression in unguided organoids at week 8 after plasmid introduction on day 11. At that time, we also did not have a functional PDGFRA antibody. Attempts to repeat the experiment did not yield additional tumors suitable for PDGFRA analysis.

We do not think that unguided organoids express inhibitory factors that block DMG transformation. Instead, we consider the limitation to be the absence of the appropriate cell populations. Indeed, it has been shown that the critical hindbrain/pons lineage required for DMG modeling is not generated in unguided organoids (Lancaster *et al.*, Nature, 2013). Further studies using our BrO system could therefore be of interest to investigate permissive cell states and to identify the factors that are critical for DMG initiation and progression.

The CAR T cell characterisation post-DMGO exposure is elegant and informative. However, a limitation is the lack of deeper analysis of resistant DMGOs post-treatment, particularly in the context of co-culture with microglia, where immunosuppressive transcriptional profiles would be expected. A more detailed exploration here would improve the robustness of the conclusions.

We are pleased that the reviewer found our functional profiling of CAR T cells in the model elegant and informative. We indeed show that in the context of microglia immunosuppressive profiles are expanded, resulting in an overrepresentation of both the exhausted CAR T cell cluster (T_{EX}) and a newly identified microglia-affected population (T_{MA}) with low effector function (**Fig. 6c,d**). However, drawing conclusions on which profiles dominate in resistant samples will require more throughput experimentation with a higher number of DMGOs, as we observed partial responses to treatment for all samples and, so far, only one sample tested displaying complete resistance to GD2 CAR T cell treatment.

Finally, in NCAM+ T cells, can the authors identify exhaustion markers, such as PI3K

pathway overactivation or epigenetic alterations, that may represent potential intervention points for small-molecule therapies—particularly those already used during CAR T cell manufacturing?

In line with NCAM1+ cells displaying a proportional increase of cells in the exhausted CAR T cell cluster (T_{EX}) (**Fig. 4I**), we find multiple exhaustion markers that define this cluster upregulated in these cells compared to NCAM1- cells (see **Figure R1**). In addition, as suggested by the reviewer we also see increased expression of PI3K in these cells (**Figure R2**). Thus, NCAM1 selection might benefit from a combined treatment approach targeting these exhaustion features to prolong the tumor-targeting efficacy of these cells. This represents an ongoing project within our team aimed at exploring strategies to overcome exhaustion and enhance engineered T cell function.

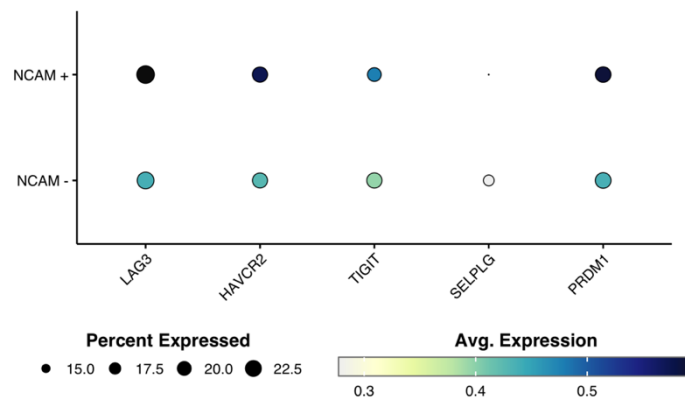


Figure R1. Exhaustion marker expression in NCAM1+ and NCAM1- cells.

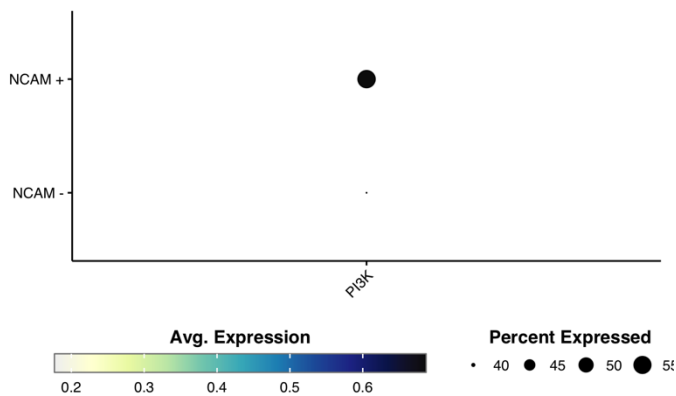


Figure R2. PI3K expression in NCAM1+ and NCAM1- cells.